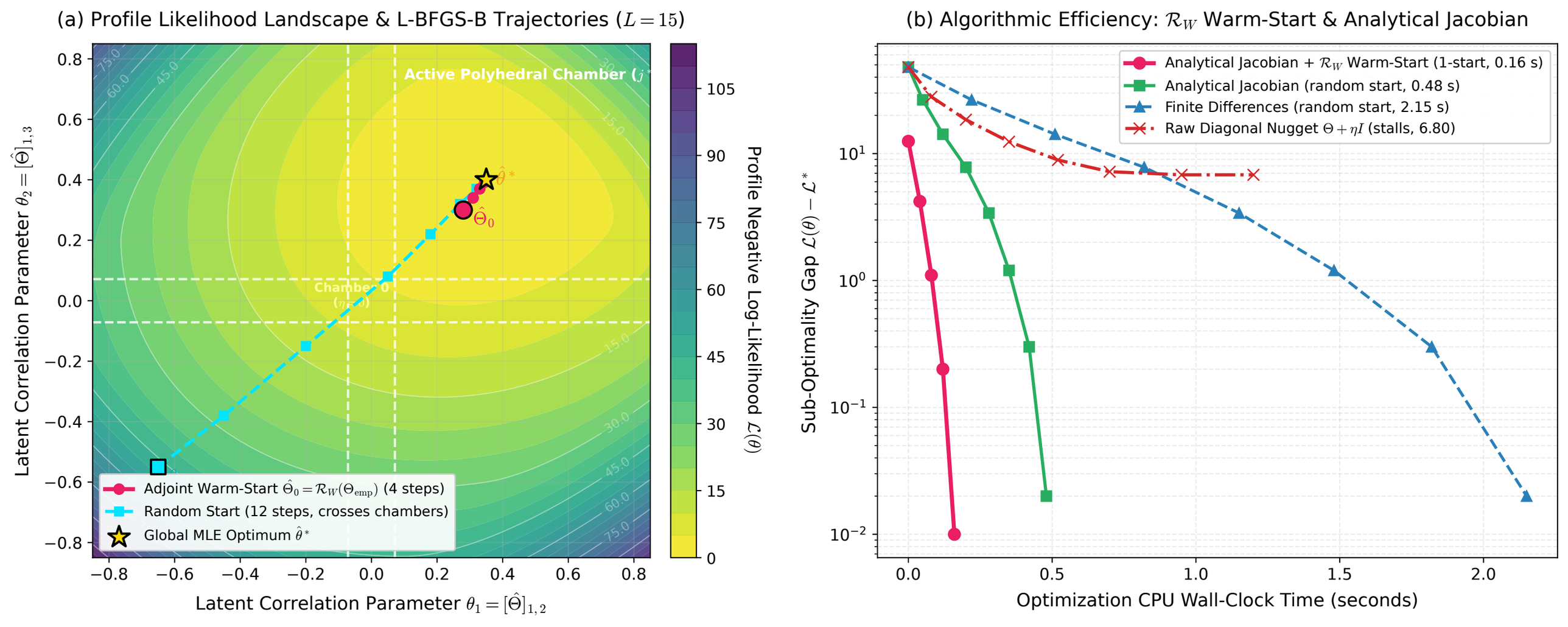

(a) Profile Likelihood Landscape & L-BFGS-B Trajectories ($L = 15$)
Active Polyhedral Chamber (j^*
Chamber 0
($\eta_{j^*} > 0$)
$\hat{\theta}^*$
$\hat{\Theta}_0$
Adjoint Warm-Start $\hat{\Theta}_0 = \mathcal{R}_W(\Theta_{\text{emp}})$ (4 steps)
Random Start (12 steps, crosses chambers)
Global MLE Optimum $\hat{\theta}^*$
Latent Correlation Parameter $\theta_1 = [\hat{\Theta}]_{1,2}$
Latent Correlation Parameter $\theta_2 = [\hat{\Theta}]_{1,3}$
Profile Negative Log-Likelihood $\mathcal{L}(\theta)$
(b) Algorithmic Efficiency: $\mathcal{R}_W$ Warm-Start & Analytical Jacobian
Analytical Jacobian + $\mathcal{R}_W$ Warm-Start (1-start, 0.16 s)
Analytical Jacobian (random start, 0.48 s)
Finite Differences (random start, 2.15 s)
Raw Diagonal Nugget $\Theta + \eta I$ (stalls, 6.80)
Sub-Optimality Gap $\mathcal{L}(\theta) - \mathcal{L}^*$
Optimization CPU Wall-Clock Time (seconds)